Shaheed Zulfikar Ali Bhutto Institute of Science & Technology

COMPUTER SCIENCE DEPARTMENT

Total Marks: 4

Obtained Marks: ________

Introduction to Data Science

Assignment # 03

Project Title:

AQI Data Science Project (KNN, Naive Bayes, K-Means & PCA)

GitHub Repository Link: https://github.com/Zainmalik3618/IDS-Assignment-3.git

Last date of Submission: 12th May, 2026

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AQI Data Science Project Report

35. Introduction

Air Quality Index (AQI) is an important indicator used to measure how polluted the air is. This project analyzes global AQI data using machine learning techniques such as K-Nearest Neighbors (KNN), Naive Bayes, K-Means Clustering, and Principal Component Analysis (PCA).

The main goal is to classify air quality levels, identify pollution patterns, and compare different machine learning models.

36. Dataset Description

The dataset contains global air quality information with the following features:

- Date
- City
- Country
- AQI
- PM2.5 (µg/m³)
- PM10 (µg/m³)
- NO2 (ppb)
- SO2 (ppb)
- CO (ppm)
- O3 (ppb)
- Temperature (°C)
- Humidity (%)
- Wind Speed (m/s)

Target variable:

- AQI Category (Good, Moderate, Unhealthy, etc.)

37. Data Cleaning Steps

- Checked missing values (no major missing data found)
- Converted AQI into categorical labels
- Removed or handled unnecessary transformations
- Standardized numerical features for clustering and KNN
- Applied feature scaling using Standard Scaler

38. Basic Statistics

From the dataset:

- Mean AQI: **164.64**
- Minimum AQI: **30**
- Maximum AQI: **300**
- Standard Deviation: **78.57**

Highest AQI City:

- São Paulo (Brazil) → **300 AQI**

Average AQI by Country (Top Insight):

- India → **170.58 (Highest average AQI)**
- Japan → 169.59
- Brazil → 168.76
- Egypt → 166.06
- UK → 163.85

39. Exploratory Data Analysis (EDA)

EDA revealed:

- PM2.5 levels strongly correlate with AQI
- Developing countries show higher AQI levels
- AQI varies significantly across cities and countries
- Seasonal trends can be observed in yearly AQI plots
- High pollution cities dominate AQI distribution

Key insight: PM2.5 is the most influential pollutant affecting AQI.

40. KNN Results

Three K values were tested:

- K = 3 → Accuracy: **0.2076**
- K = 5 → Accuracy: **0.2363**
- K = 7 → Accuracy: **0.25 (Best KNN Result)**

Best K value: K = 7

Conclusion:
KNN performed moderately but struggled with complex class boundaries.

41. Naive Bayes Results

- Accuracy: **0.3866**

Observation:

- Better than KNN
- Works well with probabilistic classification
- However, misclassified smaller AQI classes

42. K-Means Clustering Results

- Number of clusters used: **3**

Cluster analysis showed:

- Cluster 0: Low pollution cities
- Cluster 1: Moderate pollution cities
- Cluster 2: High pollution cities

Important insight: Cities naturally group into pollution-based clusters.

43. PCA Visualization

- PCA reduced data to 2 principal components
- Helped visualize AQI categories in 2D space
- Showed partial separation between pollution levels

Note:

- Variance explained was not explicitly calculated in the notebook.

44. Final Comparison Table

Model	Accuracy
KNN (Best K=7)	0.25
Naive Bayes	0.3866

Best Model: Naive Bayes performed better than KNN.

45. Conclusion

This project successfully analyzed global air quality data using machine learning techniques.

Key findings:

- India has the highest average AQI among countries
- São Paulo has the highest AQI city value
- PM2.5 is the most influential pollutant
- Naive Bayes outperformed KNN
- K-Means clustering revealed clear pollution groups
- PCA helped visualize AQI distribution patterns

46. GitHub Repository Link

https://github.com/Zainmalik3618/IDS-Assignment-3.git

47. References

- Scikit-learn Documentation
- Pandas Documentation
- Matplotlib & Seaborn Guides
- Global Air Quality Dataset (Provided in assignment)

Questions & Answers

58. What is AQI?

AQI (Air Quality Index) is a measurement used to show how clean or polluted the air is and how it affects human health.

59. Why is AQI important?

AQI is important because it helps people understand air pollution levels and possible health risks.

60. Which city or country has the highest AQI in the dataset?

São Paulo had the highest AQI value in the dataset, while India had the highest average AQI among countries.

61. Which pollutant seems most related to AQI?

PM2.5 appears to be the pollutant most strongly related to AQI.

62. What cleaning steps were required?

- Checked missing values
- Converted AQI into categories
- Applied feature scaling
- Selected important numerical features

63. What patterns did you observe from the charts?

- Countries with high pollution had higher AQI
- PM2.5 increased with AQI
- Pollution levels formed visible clusters
- AQI distribution was uneven across countries

64. Which algorithm performed better: KNN or Naive Bayes?

Naive Bayes performed better because it achieved higher accuracy than KNN.

65. What did the K-Means clusters show?

K-Means showed groups of cities with low, moderate, and high pollution levels.

66. What did PCA help you understand?

PCA helped visualize the dataset in two dimensions and showed separation between AQI groups.

67. What are the limitations of your analysis?

- Limited dataset size
- Some classes overlapped
- Accuracy could improve with advanced models
- PCA reduced some information during dimensionality reduction